

[DC] Designing and Optimizing Daily-wear Photophobic Smart Sunglasses

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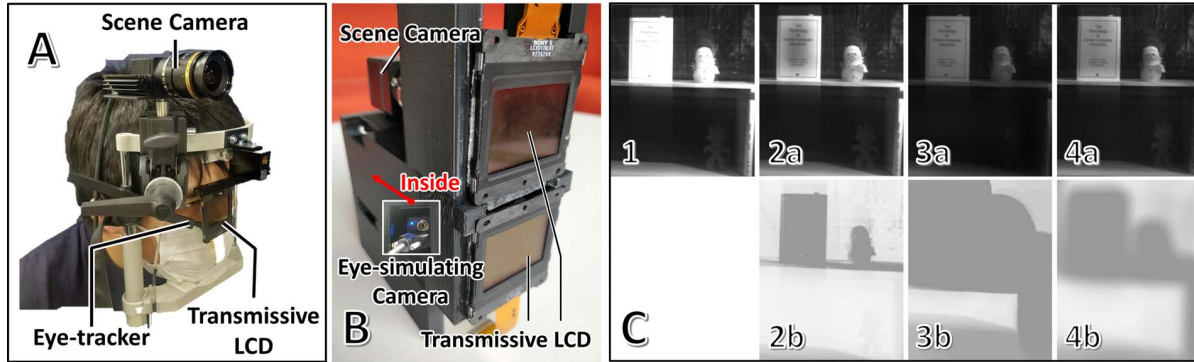


Figure 1: (A) Experimental prototype of the smart sunglasses. (B) Benchtop monocular prototype. (C) Experimental comparison: (1) The photophobia-like (overexposure) view captured by the eye-simulating camera of (B). (2a) The same observation of (1) with the unprocessed occlusion mask (2b); (3a) The same observation of (1) with the aperture-based expanded occlusion mask (3b); (4a) The same observation of (1) with the proposed optimized occlusion mask (4b).

ABSTRACT

Photophobia, also known as light sensitivity, is a condition in which there is a fear of light. Traditional sunglasses and tinted glasses typically worn by individuals with photophobia only provide linear dimming, leading to difficulty to see the contents in the dark region of a high-contrast environment (e.g., indoors at night). This paper presents a smart dimming sunglass that uses a spatial light modular (SLM) to flexibly dim the user's field of view based on scene detection from a high dynamic range (HDR) camera. To address the problem that the occlusion mask displayed on the SLM becomes blurred due to out-of-focus and thus cannot provide a sufficient modulation when viewing a distant object, I design an optimization model to dilate the occlusion mask appropriately. The optimized dimming effect is verified by the camera and preliminary test by real users to be able to filter the desired amount of incoming light through a blurred mask.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Accessibility—Accessibility systems and tools

1 INTRODUCTION AND RELATED WORK

Photophobia is an intolerance to light that is induced by a variety of ophthalmic and neurologic conditions. The way to treat photophobia is to start with the root disease that triggers photophobia, such as medication for migraines and drops to relieve inflammation for scleritis. However, some root causes of photophobia cannot be alleviated by medication, for example, some autism spectrum disorders due to large pupil size and slow pupil contraction. These types of photophobia can only be relieved by external aids, i.e., avoiding exposure to bright light and keeping the naked eye in the dark environment or wearing dark or tinted glasses. On the other hand, it is unavoidable

not to go outside even with photophobia for regular everyday life, and wearing dark or tinted glasses for a long time can make the eyes tend to be dark-adapted and produce more acute light sensitivity discomfort.

The auto-dimming eyeglasses [4] equipped with a light sensor can automatically alter the lens transmittance in response to light intensity; however, individuals with photophobia are also bothered by fluorescent lights at night, and automated global dimming glasses may obscure dark objects causing loss of visual information.

In recent years, technologies of augmented reality (AR) have been widely used to assist and treat eye diseases. Occlusion-capable AR displays [2, 5] introduce a selective light-blocking mechanism, creating an image-forming occlusion mask on an SLM between the real scene and virtual objects to alleviate the effect of translucent ghosting of virtual contents. However, to achieve pixel-by-pixel occlusion, the parallel incident light must be focused on the SLM by a convex lens. Otherwise, the occlusion mask will be out of focus and blurred as the user looks at the distant object, thus not contributing sufficient occlusion. With such an optical design, the entire system would inevitably grow bulky, and the addition of convex lenses also reduce the field of view (FOV). Meanwhile, Nayar et al. developed an adaptive dynamic range (ADR) camera [3] by equipping an SLM in front of a regular camera and adaptively adjusting the transmittance of matching pixels on the SLM panel to vary the exposure of pixels on the image detector. However, the adaptive correction via digital image data acquisition is not applicable for the eyeglasses.

Inspired by occlusion-capable AR displays and the ADR camera, I propose a photophobic smart sunglasses [1] based on a single SLM which can attenuate incident light and reduce contrast by displaying an appropriate occlusion mask on the SLM through the camera's detection of the scene. Nevertheless, as previously stated, extra optics were required to place the occlusion mask on the focal plane. Such a strategy is not practical for the everyday use of photophobic glasses. A simple geometric calculation yields that an aperture-sized occlusion mask needs to be formed in front of the eye to completely obscure a pixel in the distance. Previous studies have therefore dilated the mask in proportion to the aperture radius, this

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approach in turn causes occlusion leak (shown in Fig. 2), i.e., the mask leaks beyond the edges of the masked object. Itoh et al. offered a compensating method [2] that overlay a video see-through image onto the dilated occlusion mask to tackle this problem, however this method is ineffective for our optical see-through system. Thus, I design an optimization model to compute the optimal dilation ratio.

2 SYSTEM ARCHITECTURE AND METHOD

2.1 System Architecture

I built a benchtop prototype to confirm the proposed method and present the effect. The system structure of benchtop prototype is shown in Fig. 1(B). I also created an experimental prototype (shown in Fig. 1(A)) for user experience and installed an eye-tracker to perform an automated calibration.

The basic structure of the prototype mainly consists of a transmissive liquid crystal display (LCD) as an SLM and an HDR camera for scene detection. The real scene is captured by the scene camera and converted into a digital image, which is utilized as input to compute an occlusion mask. The mask is then displayed on the LCD to attenuate the incident light and balance the contrast of the entire scene. Thus, the user will finally see a more comfortable and less irritating scene through the LCD. In the benchtop prototype, I use a camera to simulate the human eye; the reason for not introducing a beam splitter to allow the user and the eye-simulating camera (EC) share the same viewpoint is due to the fact that the beam splitter would reduce the FOV. In addition, a secondary LCD which is without pattern dimming is placed in front of the scene camera to guarantee two cameras captured similar images since slight diffraction is generated from the gap of the liquid crystal element. And in the experimental prototype, the secondary LCD is removed because the extent of the diffraction is much less visible to the human eye than the camera.

2.2 Control Method

In my previous work [1], I briefly described the modulation function based on the provided dimming requirements. By specifying the light intensity obtained by the camera after attenuation, the transmittance of the corresponding pixel on the LCD is then calculated.

As mentioned in Sect. 1, the occlusion mask displayed on the near-eye LCD would be blurred due to being out of focus and thus cannot block the incident light effectively. On the other side, expanding the mask by the aperture radius can produce occlusion leak, obscuring the surrounding information. Therefore, I optimize the magnification of the expansion so that the pixels at the center of the mask that are not highly degraded form an effective occlusion to block the target scene; in contrast, the blurred outer part does not block excessive incident light even if it leaks out.

I first extract the blurred occlusion mask from the frame captured by the EC and simulate the blurred mask with a point spread function. Based on the brightness of the real scene, I then assign each pixel of the simulated mask a weight representing the occlusion efficiency, and if there is an occlusion leak, the weight of the associated pixel is a negative number. As a result, an expansion ratio generates an occlusion mask, and the sum of the weights determines the occlusion efficiency of the entire mask. The optimal expansion ratio is that of the mask which produces the maximum value.

3 CURRENT PROGRESS

Since this is my master's research project, this research has already made some progress with experimental results. Up to now, as shown in Fig. 1(C) and Fig. 2, it can be observed by the EC that the proposed method outperforms the global dimming approach with the unoptimized occlusion mask and the roughly expanded mask with aperture radius.

I am currently implementing eye-tracking feature with the experimental prototype shown in Fig. 1(A). The first step is to calibrate the entire system. I plan to shift the occlusion mask according to

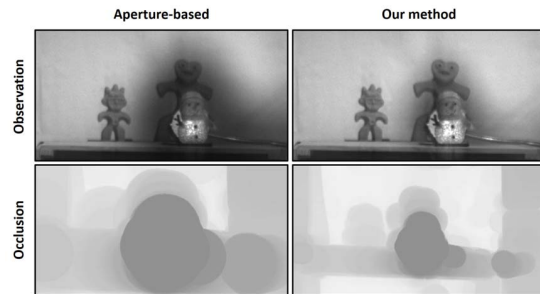


Figure 2: Occlusion leak visualization. By displaying a static occlusion mask on the LCD, the occlusion leak is made evident when the background is replaced with a light color.

the position of the pupil. By placing the EC at various positions, the corresponding projective transformation from the real scene to the observed scene can be calculated, while this projective transformation can approximately replace the one when the pupil is at the same position. Finally, the projective transformation of arbitrary positions would be obtained by interpolation.

4 DISCUSSION AND CONCLUSION

Since some photophobia is incurable, I plan to develop the smart sunglasses that can be used for daily wear by photophobic individuals. This system utilizes an SLM to selectively filter incident light based on camera detection, and shift the location of the occlusion mask in real-time by capturing the position and size of the pupil with an eye-tracker. So far, I have proposed algorithms designed for photophobia to control the SLM, the effectiveness of which can be confirmed by the camera system. However, due to the COVID-19 pandemic, user study for patients remains a non-trivial problem. Therefore, a user study conducted with individuals without photophobia is also considered, but I am still concerned that photophobic individuals may perceive the stimuli differently. In addition, I am also exploring whether there is a more suitable optimization or control method for the SLM, such as showing the deconvolved occlusion mask so that the mask can be restored to the unprocessed one after out-of-focus. However, the effect of this optimization is still needed to be verified.

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